# MISSION REPORT

## Budapest I

| | |
|---|---|
| **Status:** | ONGOING |
| **Launch Date:** | Year 1, Day 401 |
| **Operator:** | ESA |
| **Target:** | Moho |

*This document contains technical specifications and flight data for the Budapest I mission as part of the Kerbin Expansion Initiative.*

April 15, 2026

## Mission directory

## Mission objectives

The primary goal of the mission was confirming the theories about a Hohmann transfer to Moho in a reasonable time with the $\Delta$v budget of an Ariane 7 UH booster. Moho is well known for being hard to get to, especially because of its insertion burn. As such, Budapest I was tasked with finding out if it's possible to use the Ariane 7 UH to make a one-way trip to Moho from Kerbin in an optimal launch window. As a reminder, a launch window between Kerbin and Moho opens when the phase angle is about 110° (i.e. Moho is 110° ahead of Kerbin).

A secondary mission objective was testing the capabilities of the crew to transport and assemble a rover made for low-gravity in-situ. That is, the individual pieces of the rover would be stowed away in a storage container and assembled on-site by an engineer and subsequently used for excursions further away from the landing site if needed. Testing such a rover and certifying it would be very useful for future missions to Dres and Duna, even though they have higher gravity than Moho.

Although the primary goal is to test the one-way capabilities of the booster, a subsequent mission is to be organized to get the Kerbals back to Kerbin.

To summarize:

- Certify that the Ariane 7 UH booster has enough $\Delta$v for a one-way trip with a lander attached.
- Verify that the it is possible to travel and land to Moho in the established launch window with a Hohmann transfer.
- If the upper cryogenic stage has any fuel left after the insertion burn, keep it in orbit for later rendez-vous with the lander.
- Certify the rover can be built and that it works well in low-gravity situations.

More details about the rover can be found in the document detailing its performance, although a short version can also be found here.

## Kerbonauts

Because this mission is more of an engineer challenge and fixing systems on the fly is very important, two engineers and a pilot are going to be on board of Budapest İInitially the mission was designed to consist of a lander and a command module that will stay in orbit. However, because the $\Delta$v margin is slim, it is possible that a command module might not have enough fuel and the fuel aboard the lander will be needed to finish circularization, and the command module will be jettisoned. As such, it was decided that there will be an engineer left on top of the command module and one would go down to the surface, if a command module is to be included.

In the late stages of mission planning, the requirement for the command module was dropped to save weight and thus $\Delta$v, but the original requirement of two engineers remained: one to stay with the lander at all times and one to go with the rover to assist with repairs.

The team consists of one veteran of the space program and two novices:

- Dezer Kerman — Mission commander and engineer
- Urbas Kerman — Pilot, previously on Mun missions
- Bartburry Kerman — Engineer

## Vehicle specifications

Presented next are the booster and the lander configurations used for the mission. Some information is excluded here for the sake of brevity but may be found in the description documents for the individual parts, namely the booster and the MEL.

### 3.1 Ariane 7 UH booster

It is a standard Ariane 7 UH booster. The core stages are kept intact. Even though the booster has addons for refueling in orbit and airbrakes for landing, they were not removed in preparation for the mission. The payload was the MEL.

Below are the engineer readouts for the performance of the booster given the payload present:

Table 1: Ariane 7 UH Stage Specifications

| Stage | Body | Mass (t) | ISP (s) | Thrust (kN) | TWR (Max) | $\Delta v$ (m/s) |
|---|---|---|---|---|---|---|
| S2 | Moho | 69.0/108.141 | 355.0 | 750 | 2.57 (7.10) | 3,538 |
| S3 | Moho | 141.735/249.876 | 340.0 | 2,000 | 2.97 (5.45) | 2,031 |
| S4 | Kerbin (Vac) | 294.839/544.715 | 305 | 4,050 | 0.76 (1.34) | 1,695 |
| S5 | Kerbin (ASL) | 695.27/1,239.98 | 230.9 | 15,646.6 | 1.29 (2.52) | 1,528 |

A total of 8792 m/s (excluding the lander) was the total for this payload. The capabilities of the lander were excluded from the table (stages S1 and S0).

### 3.2 Moho Exploratory Lander (MEL)

The DEL was constructed with very generous fuel margins. The amount of Δv needed to safely touch down on the surface of Moho is generally considered to be about 870 m/s from a circular orbit at 20 km. The MEL however has a total amount of 3203 m/s in a vaccum (as Moho has no atmosphere), even taking into account the weight of the rover. This was done in case the upper stage did not have enough fuel to completely circularize during the capture burn, which was indeed the case. Fuel was also added to accomodate future missions, which may include bigger and heavier rover pieces or a trip to the Mohole, which would greatly increase the Δv needs. It also gives the crew the flexibility to land on the surface from a higher altitude in case a smaller periapsis could not be achieved.

It is equipped with 4 Mk-1H Torch liquid engines and RCS capabilities. The RCS was added so that the crew will be able to control its attitude and make fine adjustments when docking with the future mission meant to get them home.

The MEL has a storage capacity of 2500 liters, two solar pannel arrays and 150 units of electricity. Electricity was not of major concern, since the lander will never be piloted remotely. Of bigger concern was the amount of monopropellant in it, for which 2 more tanks of 20 units each were added to the sides.

It also presents with a Clamp-O-Tron docking port on the forward side and LT-2 landing legs.

### 3.3 Columbus

Rover parts were stowed in the cargo storage unit, along with some redundancies and 3 EVA repair kits for repairs on Columbus or the MEL. Although the most important part of the rover did not have a spare (the probe computer), it would have almost doubled the weight requirements of the cargo unit, but it would have made it possible to construct an entirely new rover with the redundancies given.

The rover was tested on Kerbin and initial testing proved it handled quite well, but the reduced gravity on Moho would potentially be a challenge to maneuverability. Upon arrival on Moho, some calibration had to be done. The final settings were reported as follows:

- Brakes on the back wheels set to 100%
- Brakes on the front wheels set to 85%
- Friction control on the back wheels set to Override, 8.0

A radioisotope electric generator was also added last minute as a way to cope with the long nights on Moho, since the solar day is 2,665,723.4 seconds.

## Trajectory

The initial trajectory was calculated to be about 7000 m/s, leaving just over 1000 m/s to spare, not taking into account the fuel left into the MEL. Unfortunately, the insertion burn was more expensive than initially assumed, along with all the corrections needed on top of the estimate from the Lambert solver.

A summary of the expenditure for every part of the mission is as follows:

| **Phase** | **Required $\Delta$v** | **Actual $\Delta$v** | **Efficiency** |
|---|---|---|---|
| Kerbin ascent | 3400 m/s | 3890 m/s | 87.4% |
| Trans-Moho injection | 1,697 m/s | 1,780 m/s | 95.33% |
| Plane change course correction | 863 m/s | 1,100 m/s | 78.45% |
| Insertion burn | 3,285 m/s | 4,000 m/s | 82.13% |
| Landing | 850 m/s | 890 m/s | 95.5% |

The crew had to use biological means for storing these numbers, and they may not be accurate. Further details on the exact trajectory is also similarly missing.

Landing occurred on day 110, year 2, at approximately 4:00 UT.

## Post-mission analysis

Talk about the conclusion of the mission, describe what the failures were, what we can learn from this, this would also be the place to write a report about incidents or kerbal-error.

## Appendix: Mission imagery

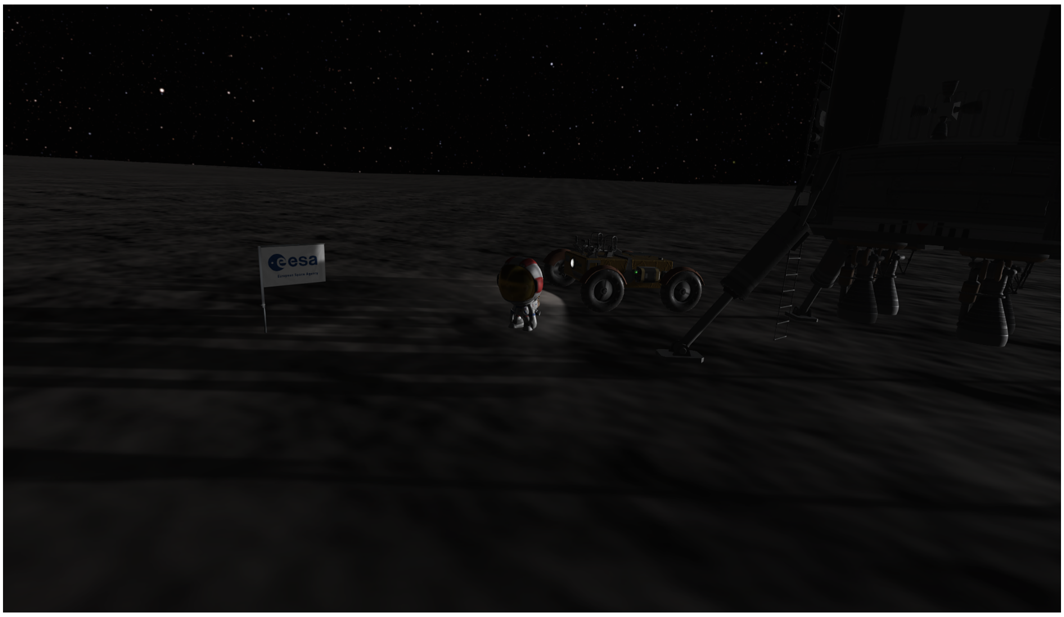


Figure 1: From left to right: the ESA flag, Dezer Kerman, the Columbus rover and the DEL lander in Dezer Crater, shortly after landing.

Figure 2: Testing the Columbus rover on Kerbin without the antenna and the radioisotope electric generator.